MZUZU UNIVERSITY

FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY (GENERIC)

SUPPLIMENTARY EXAMINATION

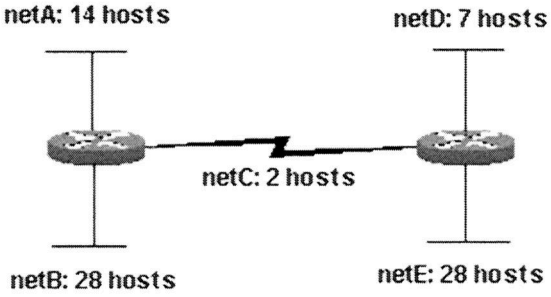
ICT 4703: NEWORK ADMINISTRATION AND INFORMATION SECURITY

TIME ALLOWED: 3 HOURS **DATE: 3rd MAY, 2018**

INSTRUCTIONS

i. This paper contains **ONE** page. Please check.
ii. There are **EIGHT** questions in this paper. Please check.
iii. Answer **ALL** questions.
iv. Marks for each question are indicated.

1. Describe any FIVE the roles of a network administrator. **[10 Marks]**
2. Describe OSI Model in relation to https communication. **[10 Marks]**
3. Explain any FIVE reasons why most systems are not secure. **[10 Marks]**
4. Explain any FIVE stages of an attack. **[10 Marks]**
5. Given the Class C network of 204.15.5.0/24, subnet the network in order to create the network in the figure below without address space wastage, with the host requirements shown. (Note: Please follow the order of networks: A, B, C, D, E) **[15 Marks]**



6. ABC Company Limited has a peer to peer network installed with an internet connection from TNM Private Liability Company. Over the years they find out that the network configuration is lacking in some crucial areas as far as information management and security is concerned. You are hired to do the advisory work and implement your ideas to come up with a highly reliable and usable network infrastructure. As a network professional with up-to-date knowledge, write the technical proposal to the ABC Company Limited on the new scalable network and the associated hardware and Software that would solve their current problems and cushion them for the future problems.

   [20 Marks]

7. Describe the RSA key generation Algorithm and use it to encrypt the Message, $M=9$ *given that* $p=7$, $q=13$ *and* $e=5$. **[15 Marks]**

8. Explain how Cryptography achieves the security goals. **[10 Marks]**

END OF QUESTION PAPER